

Hassan Ali Alvi

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EDUCATION

Ghulam Ishaq Khan University of Engineering Sciences and Technology (GIKI)

2021-2025

Bachelor of Science in Data Science

Key Course: Data Structures, Data Warehousing, Database Management Systems, Operating Systems, Computer Networking, Object Oriented Programming, Software Engineering, Data Visualizations, Data Algorithms, Data Engineering, DevOps, Data Network and Security, Parallel Processing

Aitchison College Lahore

2021

O Levels & A Levels (Major Subjects: Mathematics, Physics, Computer Science)

WORK EXPERIENCE

Future Watch

Jun 2025 – Current

- Developed and deployed Android applications with Google Play Billing, AdMob, Firebase Auth, Firestore Real-Time Sync, and cross-device consistency by **30%** and reduced data loss.
- Built optimized UI/UX for using lean back and custom layouts, improving performance and load times by **25%** and enhancing user engagement.
- Delivered full-stack mobile solutions with API Integrations, secure cloud data flows, and automated releases, reducing development turnaround by **35%** and boosting deployment efficiency.

Confiz Limited

Jun – Aug 2024

- Engaged in diverse AI and ML projects, enhanced AI model accuracy by **15%** leading to more precise predictions and improved decision-making.
- Enhanced proficiency in multiple programming languages and used tools like TensorFlow and PyTorch, collaborating with cross-functional teams to integrate AI solutions into existing systems.

Lucrum

Jul - Aug 2022

- Developed proficiency in creating interactive and insightful Power BI dashboards, navigated and leveraged ERP Systems to facilitate smoother operations and enhanced organizational productivity, leading to a **20%** improvement in overall efficiency.
- This skillset directly contributed to improving business decision-making processes, resulting in an **18%** increase in operational optimization and identifying opportunities for growth and efficiency.

ADDO AI

May – Jun 2024

- Deployed AI models, predictive analytics, and automation, leading to a **15%** improvement in decision-making and **18%** in efficiency.
- Processed and optimized large-scale datasets (100K+ records) for AI training, improving model efficiency and performance by **20%**.

Netsol Technologies

Jun - Aug 2021

- Conducted market analysis for a client to identify business opportunities with a potential of **15%** increase in market share by tapping into new revenue streams and trends.
- Created and presented business proposals, leading to a **20%** increase in client acquisition and a **25%** improvement in client retention, enhancing the company's reputation and partnerships.

Shaukat Khanum Cancer Memorial Hospital

Jun - Jul 2020

- My contributions to these efforts led to an increase in funding, enabling the organization to expand its project and initiatives, and an increase in sponsor retention, strengthening long-term partnerships and financial stability.

ACADEMIC PROJECTS

AGRO SCAN (Final Year Project):

- An AI-powered IoT system for agricultural monitoring and water management, integrating soil moisture, air, humidity, and temperature sensors to collect real-time data.
- Leveraged machine learning models to analyze and optimize farming practices, enhancing land productivity, resource efficiency (**25%** water savings), and sustainability. **(SPONSORED BY Higher Education Commission of Pakistan)**

Hey Doctor:

- Developed 'Hey Doctor,' a user-friendly application for disease diagnosis, with a comprehensive database of symptoms and treatments, enhancing patient awareness and healthcare decision-making.
- Simplified the GUI and improved accessibility to improve user satisfaction and engagement.

Social Media Engagement Analysis:

- Developed a machine learning based system to analyze sentiment in social media tweets using supervised learning models.
- Utilized KNN, Naïve Bayes, Random Forest, and Decision Trees classifiers to achieve an accuracy of **85%** in sentiment classification.

ATM Transaction Analysis:

- Developed a Power BI dashboard for comprehensive ATM transaction analysis, enabling data-driven decisions.
- Provided insights into transaction trends, peak usage times, and geographical distribution, supporting strategic planning for ATM network expansion and maintenance.

Blood Bank Management:

- Developed a Python and SQL-based Blood Bank Inventory Management System to increase operational efficiency.
- Optimized SQL commands, reducing response times, and improved decision-making in critical healthcare situations.

EXTRACURRICULARS

Aitchison College Chess Society – General Secretary

2020 – 2021

- Successfully registered around 450 participants at the Aitchison College Chess Tournament, leading a team of 30 members.

GIKI Science Society – Event Co-Ordinator

Feb 2025

- Managed the event hosted by *different societies* of GIKI and hosted more than 1500 participants throughout Pakistan.

HONOURS & AWARDS

- Captain of Aitchison College Squash Team
- Best Director, Aitchison College Art Fest 2020
- Event Co-Ordinator All Pakistan Science Fair

2021

2020

2024-2025

SKILLS & INTERESTS

Programming Languages: C (OpenMp, Cuda, MPI), C++, Java, Python (Keras, SciPy, TensorFlow, PyTorch), CSS, HTML, PySpark

Database Technologies: SQL, NoSQL, MongoDB

Visualization: Power BI, Tableau

Developer Tools: Git, Docker, VS Code, Jupyter, Google Cloud Platform, AWS, Jira

Machine Learning Models: Linear Regression, SVM, Decision Trees, Random Forests, K-Means, GMM, Logistic Regression, K-NN, PCA